RaiJouKaKu (0é0¤0¸0ç0¦0«0¯)

ITEAWD Final E-Commerce Web Application — Complete Codebase & Component Guide

Comprehensive Technical Reference for Presentation, Code Review & Evaluation

PROJECT OVERVIEW & SPECIFICATIONS

- Store Theme: Japanese Lifestyle, Crafts & Delicacies (35 products across 7 categories)
- Tech Stack: React 18, Vite, React Router v6, Material-UI (MUI), Context API, LocalStorage
- Dynamic Auth & Inventory: Registration required before sign-in, initial badge in Navbar, live stock lock (Zero Database)

1. TEAM ROLE & MODULE ASSIGNMENTS

Official project division based on the submission specification

Sangalang and Alferos

Initial project setup, folder structure (src/components), routing (React Router), global state setup, Login & Order Confirmation pages. Dynamic account registration and session management (strict registration requirement before login). Final code integration, zero-error check, archive compression & submission.

Sangalang and Mailom

Home Page & Product Listing Page. Personalized member welcome greeting on Home screen. Category banners (Electronics, Shoes, Cleaning Supplies, Sports Equipment, Grocery, Clothing, Books), search, category filtering, and multi-criteria sorting logic.

Sangalang and Mangundayao

Product Details Page & Cart Page. Product view (high-res images, quantity selector, "Add to Cart"), cart management (admin stock limit enforcement, item removal, subtotal & shipping calculations).

Sangalang and Abalos

Checkout Page & About / Contact Us Page. Multi-step shipping & billing form validation, payment method selection, order summary view, company heritage background and support contact form.

Sangalang and Ceripulo

Shared reusable components (Navbar, Footer, Product Cards, Buttons, Inputs, Badges, Modals). Initial badge in Navbar tab. Global CSS layout (main.css, components.css — pure CSS, no SCSS). Responsiveness across 430px, 867px, 1280px, and 1670px.

Everyone / Integration

PDF Documentation authoring (Store overview, Sitemap, Implemented modules list, Task distribution table), live demo coordination, in-code documentation blocks, and individual Q&A evaluation readiness.

2. ARCHITECTURE, ROUTING & GLOBAL STATE

Assigned to: Sangalang and Alferos

src/main.jsx

Type: React Entry Point (.jsx) | Assignment: Sangalang and Alferos

Primary bootstrapping entry file for React 18 createRoot() mounting #root with React.StrictMode, AuthProvider, and CartProvider.

Key Mechanisms & Logic:

- Nests global providers and imports foundational main.css.

src/App.jsx

Type: Root Theme & Provider Component (.jsx) | Assignment: Sangalang and Alferos

Configures Material-UI theme with Sakura pink palette (#F48FB1, #EC407A, #FCE4EC) and CssBaseline reset.

Key Mechanisms & Logic:

- Enforces uniform button styling and supplies ThemeProvider + RouterProvider.

src/config.js

Type: Constants & Inventory Resolver (.js) | Assignment: Sangalang and Alferos

Single source of truth for site-wide constants, currency symbols, admin credentials, localStorage keys, and inventory resolver getProductStock().

Key Mechanisms & Logic:

- Maintains getProductStock(productId) ensuring the storefront quantity matches admin inventory in real-time.
- Maintains LS_ACCOUNTS_KEY (raijoukaku_accounts) and LS_USER_KEY (raijoukaku_user) for zero-database dynamic accounts.

src/routes/paths.js

Type: Route Paths Definition (.js) | Assignment: Sangalang and Alferos

Centralizes path prefixes and dynamic helper generators.

Key Mechanisms & Logic:

- Eliminates broken links and hardcoded route strings across the storefront.

src/routes/router.jsx

Type: React Router Configuration (.jsx) | Assignment: Sangalang and Alferos

Implements createBrowserRouter with code-splitting via React.lazy() and Suspense fallback to PageLoader.

Key Mechanisms & Logic:

- Organizes nested layouts: MainLayout, EcommerceLayout, AuthLayout, and wildcard NotFound.

src/context/AuthContext.jsx

Type: Authentication & Dynamic Accounts (.jsx) | Assignment: Sangalang and Alferos

Customer session management and dynamic account registration with localStorage persistence (Zero database, zero hardcoded accounts).

Key Mechanisms & Logic:

- register(name, email, password): Persists new user to raijoukaku_accounts and auto-logs in.
- login(email, password): Strictly verifies credentials against registered accounts; rejects uncreated accounts with "Account not found. You must register an account first before signing in."
- logout(): Clears session and updates all UI components via storage events.
- Exposes user, isLoggedIn, login, register, and logout.

src/context/CartContext.jsx

Type: Shopping Cart & Stock Enforcer (.jsx) | Assignment: Sangalang and Alferos

Global shopping cart state with strict real-time admin stock enforcement.

Key Mechanisms & Logic:

- addToCart(): Automatically validates available stock; blocks adding more than remaining admin stock.
- updateQuantity(): Caps quantity at max available stock; blocks incrementing beyond admin inventory.
- Exposes getItemInCartCount and getRemainingPurchasable for all components.
- Listens to storage events to auto-sanitize cart whenever admin modifies stock in console.

src/pages/Login.jsx

Type: Customer & Admin Login Page (.jsx) | Assignment: Sangalang and Alferos

Handles customer authentication, dynamic account registration, and secret 5-click Sakura flower admin trigger (JAPANISLIFE).

Key Mechanisms & Logic:

- Supports Tab 0 (Sign In) and Tab 1 (Create Account) with full validation.
- Blocks sign-in if the account has not yet been registered in the Register tab.
- Upon registration/login, automatically routes to home screen with user active session.

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src/pages/OrderConfirmation.jsx

Type: Order Receipt Page (.jsx) | Assignment: Sangalang and Alferos

Celebratory order receipt screen displaying generated Order ID (RJK-xxxx), delivery timeline (+4 days), and purchased items.

Key Mechanisms & Logic:

- Provides immediate action buttons to continue shopping.

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3. STOREFRONT, CATALOG & FILTERING LOGIC

Assigned to: Sangalang and Mailom

src/data/categories.js

Type: Category Data Model (.js) | Assignment: Sangalang and Mailom

Defines the 7 required store categories with English and Japanese bilingual titles and banners.

Key Mechanisms & Logic:

- Electronics, Shoes, Cleaning Supplies, Sports Equipment, Grocery, Clothing, Books.

src/data/products.js

Type: Catalog Data Model (.js) | Assignment: Sangalang and Mailom

Contains all 35 authentic Japanese products with 100-base units per item.

Key Mechanisms & Logic:

- 5 curated items per category with ratings, badges, and Japanese descriptions.

src/pages/Home.jsx

Type: Storefront Home Page (.jsx) | Assignment: Sangalang and Mailom

Customer-facing landing page featuring dynamic member greeting and seasonal Hero Carousel.

Key Mechanisms & Logic:

- Displays personalized member welcome card upon login ("Kon'nichiwa, [User]!").
- Featured items strictly display live stock chip and disable add-to-cart when sold out or capped.

src/pages/ProductListing.jsx

Type: Catalog & Filter Page (.jsx) | Assignment: Sangalang and Mailom

Main catalog browsing page with multi-filter checkboxes, price slider, real-time search, and sorting.

Key Mechanisms & Logic:

- Hides or badges out-of-stock items dynamically using getProductStock().

src/components/product/ProductFilter.jsx

Type: Sidebar Filter Component (.jsx) | Assignment: Sangalang and Mailom

Category checkboxes and dual-thumb price slider emitting toggle and range change events.

Key Mechanisms & Logic:

- Clean Sakura styling on form controls.

src/components/product/ProductGrid.jsx

Type: Product Grid Layout (.jsx) | Assignment: Sangalang and Mailom

Responsive CSS Grid mapping products to ProductCard across 430px, 867px, 1280px, and 1670px.

Key Mechanisms & Logic:

- Handles empty-state feedback when zero items match search/filter.

4. PRODUCT DETAILS & CART MANAGEMENT

Assigned to: Sangalang and Mangundayao

src/pages/ProductDetails.jsx

Type: Product Details View (.jsx) | Assignment: Sangalang and Mangundayao

Comprehensive item view with live stock chip, photo gallery, and strict quantity limits.

Key Mechanisms & Logic:

- Quantity stepper is strictly capped at remaining purchasable units (stock - inCart).
- Buttons turn to disabled "Sold Out" or "Max in Cart" when inventory limit is reached.

src/pages/Cart.jsx

Type: Shopping Cart Page (.jsx) | Assignment: Sangalang and Mangundayao

Cart management table with live quantity controls capped at admin stock limit.

Key Mechanisms & Logic:

- Stepper (+) is disabled when quantity reaches available admin stock.
- Shows informative warning badges ("Max units in cart") and blocks checkout if any item is depleted.

src/components/cart/CartItem.jsx

Type: Individual Cart Row (.jsx) | Assignment: Sangalang and Mangundayao

Single product row with thumbnail, stepper, subtotal, and stock limit badge.

Key Mechanisms & Logic:

- Prevents incrementing past getProductStock(item.id) and formats currency with Philippine Peso (±).

src/components/cart/CartSummary.jsx

Type: Cart Price Summary (.jsx) | Assignment: Sangalang and Mangundayao

Calculates subtotal, evaluates free shipping (threshold ±999), and computes grand total.

Key Mechanisms & Logic:

- Includes action button to proceed to checkout.

5. CHECKOUT PIPELINE & COMPANY PAGES

Assigned to: Sangalang and Abalos

src/pages/Checkout.jsx

Type: Checkout Multi-Step Flow (.jsx) | Assignment: Sangalang and Abalos

3-step checkout wizard with Stepper and rigorous address/payment validation.

Key Mechanisms & Logic:

- Validates that no cart item exceeds admin inventory before confirming order.
- Deducts purchased quantities from localStorage raijoukaku_stock upon successful checkout.

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src/pages/AboutContact.jsx

Type: About & Customer Support Page (.jsx) | Assignment: Sangalang and Abalos

Company heritage story paired with an interactive support ticket submission form.

Key Mechanisms & Logic:

- Persists inquiries to localStorage raijoukaku_tickets for admin handling.

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src/pages/AboutUs.jsx

Type: Company Heritage Page (.jsx) | Assignment: Sangalang and Abalos

Brand storytelling detailing Tokyo artisan partnerships and authenticity guarantees.

Key Mechanisms & Logic:

- Provides direct links back to catalog.

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6. SHARED COMPONENTS, UI SYSTEM & STYLING

Assigned to: Sangalang and Ceripulo

src/assets/styles/main.css

Type: Global Stylesheet (.css) | Assignment: Sangalang and Ceripulo

Foundational stylesheet with CSS variables, typography resets, and responsive breakpoints (430px, 867px, 1280px, 1670px).

Key Mechanisms & Logic:

- Strictly 100% standard CSS — no SCSS or external frameworks.

src/assets/styles/components.css

Type: Component Stylesheet (.css) | Assignment: Sangalang and Ceripulo

Component utility classes, card hover animations, and subtle gradients.

Key Mechanisms & Logic:

- Zero build warnings.

src/components/common/Navbar.jsx

Type: Global Navigation Bar (.jsx) | Assignment: Sangalang and Ceripulo

Sticky top navbar with brand logo, live search, cart badge counter, and mobile hamburger drawer.

Key Mechanisms & Logic:

- Dynamic Auth Header: Shows initial badge (e.g. Ø<ß8 J) + Logout when logged in; shows "LOGIN | REGISTER" when logged out.
- Adapts cleanly across all screen sizes with no line-wrapping.

src/components/common/Footer.jsx

Type: Global Responsive Footer (.jsx) | Assignment: Sangalang and Ceripulo

CSS Grid/Flexbox footer with brand overview, category shortcuts, customer service contacts, and bilingual copyright.

Key Mechanisms & Logic:

- Layout prevents grid item collapse.

src/components/product/ProductCard.jsx

Type: Shared Product Card (.jsx) | Assignment: Sangalang and Ceripulo

Universal product display card with real-time stock chip and stock-aware Add to Cart button.

Key Mechanisms & Logic:

- Shows "Sold Out" or "Max in Cart" when remaining inventory is exhausted.

src/components/common/Button.jsx

Type: Reusable Button (.jsx) | Assignment: Sangalang and Ceripulo

MUI Button wrapper maintaining standardized styling.

Key Mechanisms & Logic:

- Consistent brand button hierarchy.

src/components/common/Badge.jsx

Type: Reusable Badge (.jsx) | Assignment: Sangalang and Ceripulo

MUI Chip wrapper for status tags and category chips.

Key Mechanisms & Logic:

- Standardized theme colors.

src/components/common/Breadcrumb.jsx

Type: Breadcrumb Navigation (.jsx) | Assignment: Sangalang and Ceripulo

MUI Breadcrumbs showing navigational path.

Key Mechanisms & Logic:

- Enhances user orientation across deep product views.

src/components/common/Modal.jsx

Type: Reusable Dialog (.jsx) | Assignment: Sangalang and Ceripulo

MUI Dialog wrapper with close button and responsive widths.

Key Mechanisms & Logic:

- Used for popup alerts and admin prompts.

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src/components/common/RatingStars.jsx

Type: Star Rating Display (.jsx) | Assignment: Sangalang and Ceripulo

Renders fractional star ratings (0.0 to 5.0) via MUI Rating.

Key Mechanisms & Logic:

- Consistent rating UI across cards and detail pages.

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src/components/loading/PageLoader.jsx

Type: Loading Spinner (.jsx) | Assignment: Sangalang and Ceripulo

Full-screen Sakura-themed loading spinner for route transitions.

Key Mechanisms & Logic:

- Used by React Suspense.

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7. ADMIN CONSOLE & SYSTEM INFRASTRUCTURE

System Integration & Special Modules

src/components/admin/AdminDashboard.jsx

Type: Administrative Console (.jsx) | Assignment: System Integration / All Members

Full-featured back-office management console operating purely via localStorage.

Key Mechanisms & Logic:

- Tracks 35 products with editable 100-base unit inventory steppers.
- Safe dropdown actions: Set Active, Remove from Shop, Reset 100, Restock +50, Set Sold Out.
- Order Status Dropdown: Processing, Shipped from Tokyo, In Transit, Delivered, Cancelled.
- Support Queue Dropdown: Pending Review, In Progress, Resolved, Closed.
- Any inventory change in the admin instantly locks the maximum quantity in the customer storefront.

src/components/sitemap/SitemapViewer.jsx

Type: Interactive Sitemap (.jsx) | Assignment: System Integration / All Members

Visual tree parsing sitemap.js to display all pages and route states.

Key Mechanisms & Logic:

- Provides evaluators an instant architectural overview.

src/layouts/MainLayout.jsx

Type: Main Storefront Shell (.jsx) | Assignment: System Integration / All Members

Flex column layout holding Navbar, page Outlet, and sticky Footer.

Key Mechanisms & Logic:

- Ensures footer stays grounded on short pages.

src/layouts/AuthLayout.jsx

Type: Auth Shell (.jsx) | Assignment: System Integration / All Members

Distraction-free centered layout for authentication.

Key Mechanisms & Logic:

- Clean user flow.

src/layouts/EcommerceLayout.jsx

Type: E-Commerce Route Shell (.jsx) | Assignment: System Integration / All Members

Outlet container grouping all /ecommerce/* routes.

Key Mechanisms & Logic:

- Follows enterprise routing conventions.

src/pages/NotFound.jsx

Type: 404 Error Page (.jsx) | Assignment: System Integration / All Members

Gracefully catches non-existent URLs and provides a return button to the homepage.

Key Mechanisms & Logic:

- Prevents dead ends for visitors.